MICROBIOLOGY

Retroviruses & HIV/AIDS

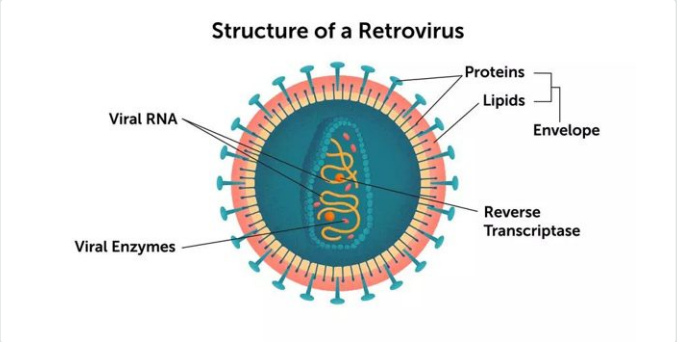
Microbiology GUE Block · MEDucation by Mattin Majid

BEFORE YOU READ — ANSWER FROM MEMORY

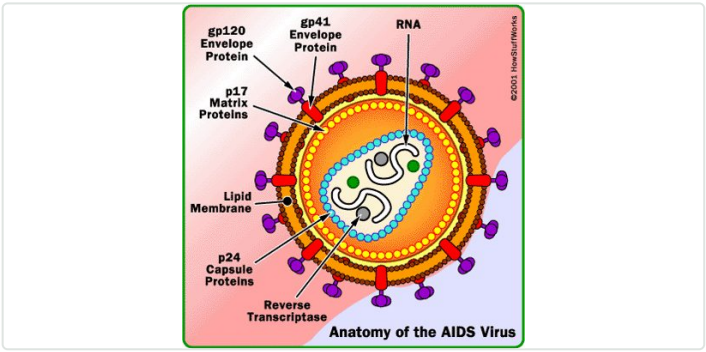
1. Which **two glycoproteins** mediate HIV binding and fusion, and which **two co-receptors** does gp120 use alongside CD4?
2. Put these in order: **reverse transcription, integration, binding, budding, fusion, assembly** — and name the enzyme responsible for each step where one is needed.
3. Why is a **negative** antibody test early after exposure NOT reassuring, and what test detects infection during this gap?

Guessing wrong before reading still improves retention — attempt all three, then check as you go.

Retroviruses — "retro" = backward; they transcribe **RNA → DNA** (reverse of the usual DNA→RNA flow), using the virally encoded enzyme **reverse transcriptase**. Structurally complex, enveloped, spherical animal viruses carrying **two identical copies of positive-sense ssRNA** plus three key enzymes within the virion: **reverse transcriptase, integrase, and a retroviral-specific protease**. Three subfamilies: **Oncovirinae** (HTLV-1, -2, -5), **Lentivirinae** (HIV-1, HIV-2), **Spumavirinae**. HIV-1 and HIV-2 cause AIDS; HTLV-1 causes adult T-cell leukemia/lymphoma (ATLL) and tropical spastic paraparesis.



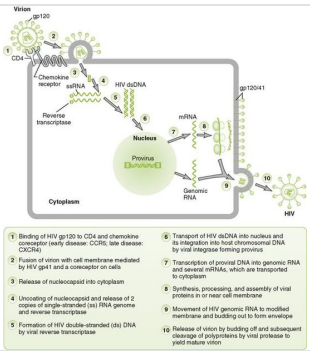
HIV virion — envelope glycoproteins, capsid, nucleocapsid, enzymes



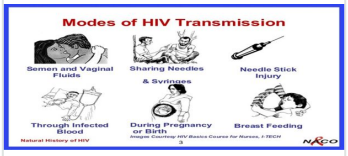
HIV anatomy labelled — gp120/gp41 envelope, p24 capsid, p17 matrix

1 Organism script — HIV

Identity	**Genus:** Lentivirus · **Family:** Retroviridae · **HIV-1** (most well-studied, >38 million affected worldwide) and **HIV-2**. Former names: HTLV-III, lymphadenopathy-associated virus (LAV), AIDS-associated retrovirus (ARV). Discovered independently by **Luc Montagnier** (France) and **Robert Gallo** (US), 1983–84
Structure	**Core:** two copies of positive-sense ssRNA + reverse transcriptase, protease, integrase, double nuclease; **p24 capsid** protein forms the protective shell. **Matrix: p17** protein surrounding the capsid, maintains virion integrity, helps direct the virus to the nucleus. **Envelope:** lipid bilayer from the host cell membrane, studded with **gp120** (docking glycoprotein) and **gp41** (transmembrane glycoprotein) — essential for binding CD4/co-receptors and membrane fusion
Entry & fusion	Requires **CD4+** plus one of two co-receptors: **CCR5** (early disease) or **CXCR4** (late disease). **gp120** binds CD4 with high affinity → conformational change → co-receptor recruitment → **gp41** inserts into and fuses with the host membrane
Replication (10 steps)	**1** gp120–CD4/co-receptor binding → **2** membrane fusion (gp41) → **3** nucleocapsid release into cytoplasm → **4** uncoating, release of 2 ssRNA copies + reverse transcriptase → **5** reverse transcriptase makes HIV dsDNA → **6** dsDNA transported to nucleus, **integrase** inserts it into host chromosomal DNA forming the **provirus** → **7** proviral DNA transcribed into genomic RNA + mRNAs → **8** viral protein synthesis/processing/assembly near the membrane → **9** genomic RNA moves to modified membrane, budding begins → **10** virion release by budding, then **viral protease** cleaves polyproteins to yield a mature virion
Transmission	**Sexual** (risk raised by concurrent STDs) · exposure to infected **blood/blood products** · contaminated **clotting factors** (hemophiliacs) · sharing **needles** (IV drug use) · **transplantation** of infected tissue/organs · **mother-to-fetus** (incl. breastfeeding). **NOT** transmitted by casual contact, touching, hugging, kissing, coughing, sneezing, insect bites, water, food, toilets, or swimming pools/public baths
Pathogenesis	HIV destroys CD4+ cells by 3 mechanisms: **(1)** viral replication punches holes in the cell membrane, **(2)** infected cells fuse with others forming a **syncytium** that destroys all fused cells, **(3)** anti-HIV antibodies bind infected cells and activate **complement**, destroying them. Early post-exposure: high blood HIV level (detected by HIV antigen + HIV-RNA assays) → settles to a **set-point** during incubation, with massive CD4 turnover (killed cells replaced efficiently) → AIDS develops once **killed CD4 cells can no longer be replaced** (high HIV-RNA, high HIV-antigen, low CD4)
Clinical course	**Acute HIV illness:** most patients asymptomatic, but a glandular-fever-like illness, hepatitis, meningitic illness or thrombocytopenia can occur; high viremia at this time. **Asymptomatic chronic infection:** steady state between virus and immune system, lasts **5–15 years**, low viral load, normal CD4 count; some develop persistent generalized lymphadenopathy (PGL). **AIDS:** rising viral load + falling CD4 → opportunistic infections and immunodeficiency-associated conditions. HIV-1 and HIV-2 cause clinically indistinguishable illness; patients are **infectious throughout life**, highest during acute illness and full-blown AIDS



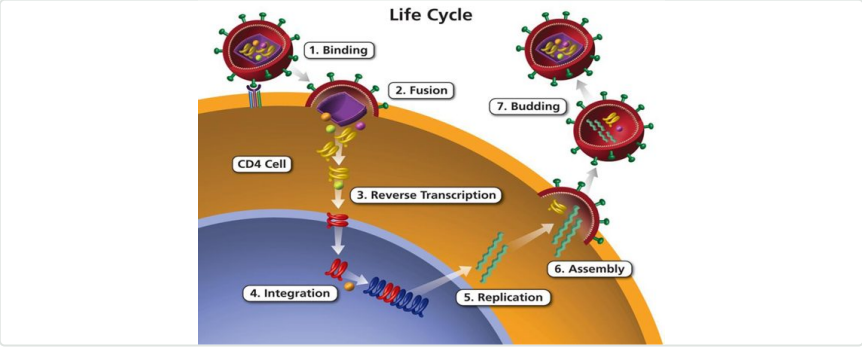
HIV replication cycle, 10 labelled steps — binding through mature virion release



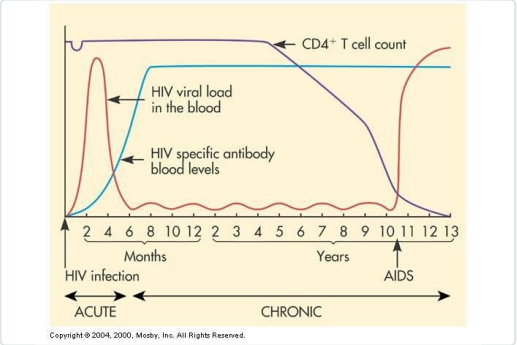
Modes of HIV transmission

2 Immunologic manifestations over time

Stage	Finding
Early (window period)	Slight, temporary CD4 depression, few symptoms; **antibody window up to 6 weeks** (by 6 months, 95% positive); during the window, **p24 antigen** is present with acute viremia/antigenemia
Antibody sequence	Antibodies form to all major antigens; **first detected against gag proteins p24 and p55**, followed by antibody to p51, p120 and gp41; antibody levels **decrease as disease progresses**
Cellular changes	**CD4 falls** — budding, syncytial fusion, apoptosis; B cells show decreased antigen response (viral proteins may block T–B cell interaction at the CD4 site); **CD8 initially rises** and may stay elevated; as CD4 drops, immunosuppression and susceptibility to opportunistic disease follow



HIV life cycle, summary view — binding → fusion → reverse transcription → integration → replication → assembly → budding



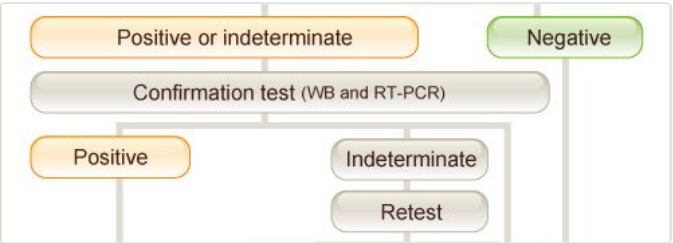
Viral load, CD4 count, and antibody levels across acute → chronic → AIDS

3 Diagnosis

Test	Role / interpretation
ELISA	**Initial screening**; most reliable screening test. **Sensitivity >99.9%**; 50% positive by <22 days, 95% by <6 weeks. Two different ELISA results must be positive before confirmatory testing. P24-antigen ELISA aids **early** detection. Needs Western blot confirmation
Western blot	**Gold-standard confirmatory test** — specificity >99.9% when combined with ELISA. Viral antigens separated: **gp160, gp120, p66, p55, p51, gp41, p31, p24, p17, p15**; interpretation depends on which bands react
RT-PCR	Detects **viral RNA** in blood directly
CD4:CD8 ratio	Used for **staging** the disease
Rapid tests	Growing importance; results in **<20 minutes**, less technical skill required, but still need ELISA/Western blot confirmation
Viral load tests	Measure actively replicating virus; correlate with response to antiretrovirals; **most helpful before seroconversion**. Higher baseline viral load → higher 5-year progression rate to AIDS
Negative antibody result	Means either HIV-negative **or** too early for seroconversion (recent infection) — CDC recommends **follow-up testing at 6 weeks, 12 weeks, and 6 months**



Western blot strips — band pattern determines interpretation



HIV testing algorithm — screening → confirmation → result

4 CD4 lymphocyte count — clinical use

CD4 count	Interpretation
800–1200/µl	Normal
>500/µl	Generally healthy

200–499/µl	Minor immune problems (problems begin below 500/µl)
<200/µl	Severe problems — high risk of AIDS-defining illness, opportunistic infection, or malignancy
CD4% <20%	Watch for opportunistic infection/malignancy onset; **CD4:CD8 ratio reverses** from 2:1 to 1:2

P Must-remember pearls

- **gp120 binds CD4** (docking); **gp41 drives fusion** — know which glycoprotein does which job.
- The **window period** (up to 6 weeks, 95% seroconverted by 6 months) is why a single negative antibody test early after exposure doesn't rule out infection — **p24 antigen or RT-PCR** catch this gap.
- **ELISA screens, Western blot confirms** — never diagnose HIV on ELISA alone; two positive ELISAs are needed before confirmatory testing even begins.
- **CD4 <200/µl** is the single number to remember for severe immunosuppression and AIDS-defining illness risk.
- HIV destroys CD4 cells by **three mechanisms**: membrane damage from replication, syncytium formation, and antibody/complement-mediated killing — not simple direct lysis alone.

Q Test yourself

1. A patient has a high-risk sexual exposure 3 weeks ago and an HIV antibody ELISA returns negative. What is the most appropriate next step?

A. Reassure the patient they are HIV-negative with no further testing
B. Repeat antibody testing at 6 weeks, 12 weeks and 6 months, since this may be the window period
C. Diagnose AIDS based on the negative result
D. Order a CD4:CD8 ratio only
E. No further action is needed since ELISA sensitivity is 100% at any time point

2. A patient's HIV Western blot shows reactivity to gp120, gp41 and p24. What is the correct interpretation of this test in the diagnostic pathway?

A. This is the initial screening test
B. This is the gold-standard confirmatory test, and this pattern supports a positive result
C. This test detects viral RNA directly
D. This test is used only for CD4 staging
E. This test result alone, without any screening test, is sufficient for diagnosis

3. A newly infected patient has a high level of HIV antigen and HIV-RNA in blood but a negative antibody test, 2 weeks after exposure. What explains this combination?

A. The patient cannot be infected since the antibody test is negative
B. This is the window period — before seroconversion but with high viremia/antigenemia detectable by antigen and RNA assays
C. The patient has progressed directly to AIDS
D. This indicates HTLV-1 infection instead
E. This indicates a false-positive antigen test

4. A patient's CD4 count drops to 150 cells/µl with a reversed CD4:CD8 ratio. What is the most appropriate interpretation?

A. Normal immune status
B. Minor immune problems only
C. Severe immune problems with high risk of opportunistic infection or AIDS-defining illness
D. This count indicates recent infection only, with no clinical significance
E. This count is inconsistent with HIV infection

5. Which combination of receptor and co-receptor interactions is required for HIV entry into a target cell, and which viral protein mediates the subsequent membrane fusion step?

A. CD8 plus CCR5, with gp120 mediating fusion
B. CD4 plus CCR5 or CXCR4, with gp41 mediating fusion
C. CD4 alone, with p24 mediating fusion
D. CD8 plus CXCR4, with p17 mediating fusion
E. CD4 plus CCR5 only, with integrase mediating fusion

ANSWERS — COVER UNTIL YOU'VE COMMITTED

1 — A negative antibody test within the window period (up to 6 weeks, with 95% seroconverted by 6 months) does not exclude infection — CDC-recommended follow-up testing at 6 weeks, 12 weeks and 6 months is needed.

2 — Western blot is the gold-standard confirmatory test; reactivity to viral antigens such as gp120, gp41 and p24 supports a positive interpretation, but it is always used after a positive screening ELISA, not alone.

3 — During the window period, antibody has not yet formed, but p24 antigen and HIV-RNA are present due to high acute viremia and antigenemia — this is why antigen/RNA-based tests are used for early detection.

4 — A CD4 count below 200/µl with a reversed CD4:CD8 ratio (normally 2:1, reversed to 1:2) indicates severe immune problems and high risk of AIDS-defining illness, opportunistic infection, or malignancy.

5 — HIV entry requires CD4 plus one of two co-receptors, CCR5 (early disease) or CXCR4 (late disease); gp120 mediates receptor binding while gp41 mediates the subsequent membrane fusion.

SPACED REVIEW

☐ Day 1	☐ Day 3	☐ Day 7	☐ Day 21	☐ Pre-exam

Microbiology GUE Block · MEDucation by Mattin Majid · Source: SGL 3 — Retroviruses and Human Immunodeficiency Virus (HIV) / AIDS (HMU College of Medicine, Microbiology/Physiology/Genetics Dept.)
Study alongside the lecture, not instead of it.